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Preface

A short preface.

PART I

The First Part

An example chapter

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1.1 Background and motivation

We set the historical and physical context.

1.1.1 The Yang–Baxter equation

A consistency condition for factorised scattering [1,2]:

$$R_{12}(u)R_{13}(u+v)R_{23}(v) = R_{23}(v)R_{13}(u+v)R_{12}(u) \tag{1.1}$$

> The Yang–Baxter equation is the single organising principle behind every construction in Part I.

1.1.2 Transfer matrices

The algebraic backbone of Part I.

1.2 Scope of this thesis

A boxed key result:

$$[R_{12}, R_{13}] + \dots = 0 \tag{1.2}$$

1.3 The reflection equation

The boundary analogue of the bulk relation, due to Sklyanin [3]:

$$R_{12}K_1R_{21}K_2 = K_2R_{12}K_1R_{21} \tag{1.3}$$

An unnumbered aside:

$$K(u)\,\overline{K}(u) = 1$$

Manual spacing:

$$\int f(x)\,\mathrm{d}x \quad \text{vs.} \quad \int f(x)\,\mathrm{d}x \tag{1.4}$$

Figure 1.1: A placeholder figure.

(a) **(b)**

Figure 1.2: Two related panels.

> **Key idea**
> At a boundary, integrability is controlled by the reflection equation rather than the bulk relation alone.

1.4 Long equations

A numbered display wrapped in `breakeq` may split across pages at its line breaks, with its number set once beside the last line:

$$\begin{aligned} f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \\ + a_4 x^4 + a_5 x^5 + a_6 x^6 \\ + a_7 x^7 + a_8 x^8 + \dots \end{aligned} \tag{1.5}$$

See Equation (1.5) and Appendix 1.A.

1.5 A worked figure

Appendix 1.A ◇ Detailed derivation of the R-matrix

Expanding the consistency condition to low order,

$$R(u) = 1 + u\, r + O(u^2), \tag{1.A.1}$$

and collecting the next order gives

$$r_{12} r_{13} + r_{12} r_{23} + r_{13} r_{23} = 0. \tag{1.A.2}$$

Appendix 1.B ◇ Conventions and notation

Traces are taken over the auxiliary space,

$$\tau(u) = \mathrm{tr}_0 T_0(u). \tag{1.B.1}$$

PART II

The Second Part

A second chapter

Equations number per chapter, and references to them stay correct from anywhere: Equation (1.5) lives in Chapter 1.

$$Z = \sum_{\{\sigma\}} \exp(-\beta H[\sigma]), \qquad \sum_{\substack{i\in\{\alpha^{(q)}\}\\ j\in\{\beta^{(p)}\}}} J_{ij}|\psi\rangle \tag{2.1}$$

Model	Symmetry	Solved by
XXX	$\mathfrak{su}(2)$	Bethe ansatz
XXZ	$U_q(\mathfrak{su}(2))$	Bethe ansatz

Coauthor contributions

All chapters of this thesis were written under the supervision of …

Chapter 1: An example chapter

Y. Name, A. Coauthor and B. Supervisor
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YN performed the calculations and wrote the text. AC contributed to the analysis. BS supervised the project.

Summary

Summary in English, bearing the title of the thesis.

Samenvatting

Samenvatting in het Nederlands, met de titel van het proefschrift.

Acknowledgements

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